

16/1/19

1:30 - 4:30

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Paper Code: TCS/TIT-405

End Semester Back Paper Examination (even), 2019

B.tech 5th sem.(CS&IT)

Theory of computation

Time: Three Hours

MM: 100

Note:

- (i) This question paper contains five questions with alternative choice.
- (ii) All questions are compulsory..
- (iii) Each question carries three parts a,b and c. Attempt any two parts.
- (iv) Each part carries **ten** marks. Total marks assigned to each question are **twenty**.

Q1. (a) Explain Finite automata. Describe the types of automata .

(b) Prove that a language accepted in DFA is also acceptable in NFA.

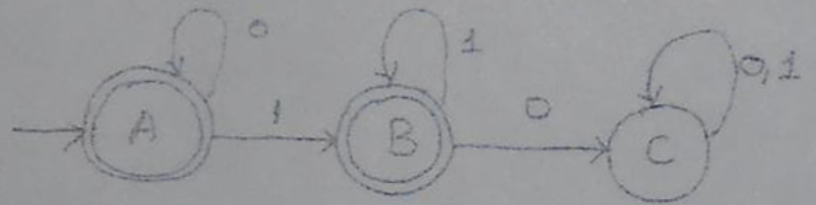
(c) What do you understand by language generator and language recognizer? Explain with example. .

Q2. (a) What is Grammar and language? Explain with example.

(2*10=20)

(b) Show that the set $L = \{0^n 1^{n+1} / n \geq 1\}$ is not a regular.(c) Construct an NFA equivalent to $(a+b)^*(ab+ba)(a+b)^*$.

Q3. (a) Obtain the regular expression that denotes the language accepted by the following DFA.



(b) Show that the regular languages are closed under intersection and reversal.

(c) State pumping lemma for Context free language with example.

Q4. (a) Let G be a grammar $s \rightarrow OB/1A$, $A \rightarrow O/OS/1AA$, $B \rightarrow 1/1S/OBB$. For the string 00110101 find its leftmost derivation and derivation tree

(b) i) What do you understand by ambiguity in CFG.

ii) If G is the grammar $S \rightarrow aS/Sa/a$, Show that G is ambiguous.

(c) Construct a PDA M accepting $L = \{a^n b^n \mid n \geq 1\}$.

Q5.(a) Explain universal Turing machine in detail.

(2*10=20)

(b) Discuss the properties of Recursive and Recursive enumerable languages.

(c) Explain halting problem in Turing machine in detail.